Notes on Axionic wormholes

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1 Action principle and equations of motion

Our starting point is the following action

$$\mathcal{S} = \int d^D x \sqrt{g} \left[\frac{1}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) + \frac{1}{2(D-1)!} H_{\mu \dots \nu} H^{\mu \dots \nu} \right] . \tag{1.1}$$

H is a $(D-1)$-form field strength satisfying the Bianchi identity $dH = 0$. Thus, one can always write locally $H = dB$ where B is a $(D-2)$-form field. The equations of motion are simply $d \star H = 0$. We can see classically that the dynamics of H is locally governed by a pseudo-scalar field χ defined as $\star H = i\, d\chi$.

That holds at the quantum level. To see this we introduce χ as a Lagrangian multiplier in the following 1st class formalism action

$$\begin{aligned} \mathcal{S}_{1st} &= \int d^D x \frac{\sqrt{g}}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) + \frac{1}{2} \int H \wedge \star H + \int \chi \wedge dH \\ &= \int d^D x \frac{\sqrt{g}}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) + \frac{1}{2} \int H \wedge \star H - \int d\chi \wedge H + \mathcal{S}_{\partial M} . \end{aligned} \tag{1.2}$$

where

$$\mathcal{S}_{\partial M} = \int_{\partial M} \chi \wedge H . \tag{1.3}$$

The variational principle is well defined if we impose Dirichlet boundary conditions for χ and for H. The path-integral whose action is defined by (1.1) can be recovered by integrating out χ imposing Bianchi's identity. In the meantime, the action is quadratic in H so we can integrate out H as well by solving its equation of motion

$$H = \star\, d\chi \Rightarrow \star H = (-1)^{D-1} d\chi . \tag{1.4}$$

Indeed, inserting back into the action we get

$$\mathcal{S}_\chi = \int d^D x \frac{\sqrt{g}}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) - \frac{1}{2} \int d\chi \wedge \star d\chi - \int_{\partial M} \chi \wedge \star d\chi . \tag{1.5}$$

Instanton configurations have the effect of modifying the action by mean of a non-perturbative potential $V_{\rm np}(\chi)$ so that the total action in the pseudo-scalar variable reads

$$\mathcal{S} = \int d^D x \sqrt{g} \left[\frac{1}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) - \frac{1}{2} \partial_\mu \chi \partial^\mu \chi + V_{\rm np}(\chi) \right] + \mathcal{S}_{GHY} + S_{\partial M} . \tag{1.6}$$

We look for homogeneous and isotropic solutions of the form

$$ds^2 = N^2(\tau) d\tau^2 + a^2(\tau) d\Sigma_d^2 \quad , \quad \chi = \chi(\tau) . \tag{1.7}$$

We obtain the following expressions for the determinant of the metric and the Ricci scalar

$$\sqrt{g} = Na^d \quad , \quad \mathcal{R} = -\frac{2d}{N}\frac{d}{d\tau}\left(\frac{a'}{aN}\right) + d(d-1)\frac{k}{a^2} - d(d+1)\left(\frac{a'}{aN}\right)^2 . \tag{1.8}$$

Finally, let's also mention the form of the Gibbons-Hawking term

$$\mathcal{S}_{GHY} = -\frac{1}{\kappa}\int_{\partial M} d^d x\sqrt{h}K = -\frac{1}{\kappa}\frac{\partial_\tau}{N}V_d(a^d) = -\frac{dV_d a' a^{d-1}}{\kappa N} . \tag{1.9}$$

Hence the Lagrangian density of the Einstein-Hilbert part of the action reads

$$\begin{aligned}\mathcal{L}_{EH} &= -\frac{Na^d}{2\kappa}\left[-\frac{2d}{N}\frac{d}{d\tau}\left(\frac{a'}{aN}\right) + d(d-1)\frac{k}{a^2} - d(d+1)\left(\frac{a'}{aN}\right)^2 + 2\Lambda\right] \\ &= -\frac{1}{2\kappa}\left[\frac{d(d-1)(a')^2 a^{d-2}}{N} + d(d-1)Nka^{d-2} - 2N\Lambda a^d\right] + \frac{d}{d\tau}\left[\frac{da^{d-1}a'}{\kappa N}\right] .\end{aligned} \tag{1.10}$$

The boundary term coming from this total derivative will get exactly canceled by the Gibbons-Hawking term (see (1.9)). We also have

$$\mathcal{L}_\chi = -\frac{a^d}{2N}(\partial_\tau\chi)^2 + Na^d V_{\mathrm{np}}(\chi) . \tag{1.11}$$

Also, note that taking the trace of Einstein equation allows one to evaluate the on-shell action as a function of the on-shell solution χ. Indeed,

$$\begin{aligned}&G = \kappa T^\chi - \Lambda D \Rightarrow \mathcal{R}\left(1 - \frac{D}{2}\right) = \kappa T^\chi - \Lambda D , \\ &T^\chi_{\mu\nu} = -\partial_\mu\chi\partial_\nu\chi - g_{\mu\nu}\left(-\frac{1}{2}(\partial_\lambda\chi)^2 + V_{\mathrm{np}}(\chi)\right) \Rightarrow T^\chi = \left(\frac{D}{2} - 1\right)(\partial_\lambda\chi)^2 - DV_{\mathrm{np}}(\chi) .\end{aligned} \tag{1.12}$$

After some manimpulations one obtains

$$\frac{-\mathcal{R} + 2\Lambda}{2\kappa} = \frac{1}{2}(\partial_\lambda\chi)^2 - \frac{DV_{\mathrm{np}}}{D-2} - \frac{2\Lambda}{\kappa(D-2)} . \tag{1.13}$$

We also have the additional boundary terms

$$\mathcal{S}_{GHY} = -\frac{dV_d a' a^{d-1}}{\kappa N}\bigg|_{\partial M} \quad , \quad \mathcal{S}_{\partial M} = -\frac{V_d}{N}\, a^d \chi\partial_\tau\chi\bigg|_{\partial M} . \tag{1.14}$$

The on-shell action thus takes the simple expression

$$S^{\mathrm{on-shell}} = V_d \int d\tau N a^d \left(-\frac{2V_{\mathrm{np}}}{D-2} - \frac{2\Lambda}{\kappa(D-2)}\right) + \mathcal{S}_{GHY} + \mathcal{S}_{\partial M} . \tag{1.15}$$

This on-shell action is an important quantity since the final question one should answer is how it evolves as a function of the number of bubbles. The flow paramater is $u = N\tau$. Remark that the on-shell action does not depend on the kinitic term and can be easily deduced from this formula once χ and the scale factor is known.

Let's now derive the equation of motions. The simplest one is obtain by varying wrt to χ yielding

$$\partial_\tau\left(\frac{a^d}{N}\partial_\tau\chi\right) + Na^d\partial_\chi V_{np}(\chi) = 0 . \tag{1.16}$$

Varying with respect to a in $\mathcal{L}_{EH}+\mathcal{L}_\chi$ yields

$$\begin{aligned}&\Bigg[-\frac{da^{d-1}}{2N}(\partial_\tau\chi)^2+dNa^{d-1}V_{\rm np}(\chi)-\frac{1}{2\kappa}\Big(\frac{d(d-1)(d-2)}{N}a'^2a^{d-3}\\&+d(d-1)(d-2)kNa^{d-3}-2N\Lambda da^{d-1}\Big)\Bigg]\delta a-\frac{1}{2\kappa}\frac{d(d-1)2a'a^{d-2}}{N}\delta(a')\,.\end{aligned}\tag{1.17}$$

Thus, integrating by parts the second term

$$\begin{aligned}&\Bigg[-\frac{da^{d-1}}{2N}(\partial_\tau\chi)^2+dNa^{d-1}V_{\rm np}(\chi)-\frac{1}{2\kappa}\Bigg(\frac{d(d-1)(d-2)}{N}a'^2a^{d-3}\\&+d(d-1)(d-2)kNa^{d-3}-2N\Lambda da^{d-1}-2d(d-1)\frac{d}{d\tau}\left(\frac{a'a^{d-2}}{N}\right)\Bigg)\Bigg]\delta a\,.\end{aligned}\tag{1.18}$$

We thus get the following equation of motion

$$\begin{aligned}&\frac{da^{d-1}}{2N}(\partial_\tau\chi)^2-dNa^{d-1}V_{\rm np}(\chi)+\frac{1}{2\kappa}\Bigg(\frac{d(d-1)(d-2)}{N}a'^2a^{d-3}\\&+d(d-1)(d-2)kNa^{d-3}-2N\Lambda da^{d-1}-2d(d-1)\frac{d}{d\tau}\left(\frac{a'a^{d-2}}{N}\right)\Bigg)=0\,.\end{aligned}\tag{1.19}$$

Finally, varying wrt to N we generate a third (non-independent) equations

$$a^dV_{\rm np}(\chi)+\frac{a^d}{2N^2}(\partial_\tau\chi)^2-\frac{1}{2\kappa}\left[d(d-1)ka^{d-2}-2\Lambda a^d-\frac{d(d-1)a'^2a^{d-2}}{N^2}\right]=0\,.\tag{1.20}$$

Let's summurize the set of equations here for general (k, Λ, N)

$$\begin{aligned}\partial_\tau\left(\frac{a^d}{N}\partial_\tau\chi\right)+Na^d\partial_\chi V_{np}(\chi)&=0\,.\\ \frac{d(d-1)}{a^2}\left(\frac{a'^2}{N^2}-k\right)+2\kappa\left(\frac{(\partial_\tau\chi)^2}{2N^2}+V_{\rm np}(\chi)+\frac{\Lambda}{\kappa}\right)&=0\,,\\ \frac{(d-1)(d-2)}{a^2}\left(k+\frac{a'^2}{N^2}\right)-\frac{2(d-1)}{Na^{d-1}}\frac{d}{d\tau}\left(\frac{a'a^{d-2}}{N}\right)+2\kappa\left(\frac{(\partial_\tau\chi)^2}{2N^2}-V_{\rm np}(\chi)-\frac{\Lambda}{\kappa}\right)&=0\,.\end{aligned}\tag{1.21}$$

In the $N=1$ gauge, it yields

$$\begin{aligned}\partial_\tau\left(a^d\partial_\tau\chi\right)+a^d\partial_\chi V_{np}(\chi)&=0\,.\\ \frac{d(d-1)}{a^2}\left(a'^2-k\right)+2\kappa\left(\frac{(\partial_\tau\chi)^2}{2}+V_{\rm np}(\chi)+\frac{\Lambda}{\kappa}\right)&=0\,,\\ \frac{(d-1)(d-2)}{a^2}\left(k+a'^2\right)-\frac{2(d-1)}{a^{d-1}}\frac{d}{d\tau}\left(a'a^{d-2}\right)+2\kappa\left(\frac{(\partial_\tau\chi)^2}{2}-V_{\rm np}(\chi)-\frac{\Lambda}{\kappa}\right)&=0\,.\end{aligned}\tag{1.22}$$

Non-running solutions A trivial set of solutions can be obtained by setting χ to a extremum of the potential. These solutions will benchmark of the more general solutions. Let $V_{\rm np}^{\rm ext}$ an extremum of the non-perturbative potential. The geometry of the solution depends crucially on the sign of $\Lambda_{\rm eff}=\kappa V_{\rm np}^{\rm ext}+\Lambda$. We work in the $N=1$ gauge.

- If $\Lambda_{\rm eff}>0$
  - If $k=1\Rightarrow a(\tau)=\frac{1}{H_{dS}}\sin(H_{dS}\tau)$, with $H_{dS}=\sqrt{\frac{2\Lambda_{\rm eff}}{d(d-1)}}\Rightarrow EdS_{d+1}$ in spherical slices.

- Otherwise, there are no solutions for a real.

- If $\Lambda_{\text{eff}} < 0$
  - If $k = 1 \Rightarrow a(\tau) = \frac{1}{H_{AdS}} \sinh(H_{AdS}\tau)$, with $H_{AdS} = \sqrt{-\frac{2\Lambda_{\text{eff}}}{d(d-1)}} \Rightarrow EAdS_{d+1}$ in spherical slices.
  - If $k = -1 \Rightarrow a(\tau) = \frac{1}{H_{AdS}} \cosh(H_{AdS}\tau) \Rightarrow EAdS_{d+1}$ in hyperbolic slices.
  - If $k = 0 \Rightarrow a(\tau) = \frac{1}{H_{AdS}} \exp(H_{AdS}\tau) \Rightarrow EAdS_{d+1}$ in flat slices.
- If $\Lambda_{\text{eff}} = 0$
  - If $k = 1 \Rightarrow a(\tau) = \tau \Rightarrow \mathbb{R}_{d+1}$. These are cylinder coordinates associated to Euclidean space.
  - If $k = 0 \Rightarrow a(\tau) = \text{const}$. Upon rescaling the transverse flat slices, these are standard cartesian coordinates.

We now focus on compact=spherical slices to obtain a finite action. We could also regulate the infinite volume of the transverse slices but we postpone this issue for latter. In these non-running solutions, the boundary term from the axion part trivially vanishes. We also observe that the bulk on-shell action always take the simple form

$$S_{\text{bulk}}^{\text{on-shell}} = -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)} \int d\tau a^d \,. \tag{1.23}$$

- If $\Lambda_{\text{eff}} > 0$: There is no boundary since the geometry is a sphere and thus the Gibbons-Hawking York term trivially vanishes. One finds

$$\begin{aligned} S^{\text{on-shell}} &= -\frac{V_d\Lambda_{\text{eff}}}{\kappa(D-2)H_{dS}^{d+1}} \frac{\sqrt{\pi}\Gamma\left(\frac{1+d}{2}\right)}{\Gamma(1+\frac{d}{2})} \\ &= -\frac{d\,V_{d+1}}{\kappa H_{dS}^{d-1}} \,. \end{aligned} \tag{1.24}$$

  This should be seen as the semi-classical contribution to the dS free energy.

- If $\Lambda_{\text{eff}} < 0$: We have an asymptotically EAdS space and thus the action should diverge when approaching the boundary at $\tau \to \infty$. We thus regulate the action by fixing the radius of the boundary $a(\tau_f) = R$

$$\begin{aligned} S_{\text{bulk}}^{\text{on-shell}} &= -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)} \int d\tau a^d = -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)H_{AdS}^{d+1}} \int_0^{H_{AdS}R} du \frac{u^d}{\sqrt{1+u^2}} \,. \\ &= -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)} \frac{R^{d+1}}{d+1} \, {}_2F_1\left(\frac{1}{2}, \frac{1+d}{2}, \frac{3+d}{2}, -(H_{AdS}R)^2\right) \,. \end{aligned} \tag{1.25}$$

  The Gibbons-Hawking yields

$$S_{GHY} = -\frac{dV_d\sqrt{1+H_{AdS}^2R^2}R^{d-1}}{\kappa} \,. \tag{1.26}$$

  Combining these two contributions yield

$$S^{\text{on-shell}} = -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)} \frac{R^{d+1}}{d+1} \, {}_2F_1\left(\frac{1}{2}, \frac{1+d}{2}, \frac{3+d}{2}, -(H_{AdS}R)^2\right) - \frac{dV_d\sqrt{1+H_{AdS}^2R^2}R^{d-1}}{\kappa} \,. \tag{1.27}$$

- If $\Lambda_{\text{eff}} = 0$: the bulk on-shell action trivially vanishes. We still need to regularize the integral because the GHY term diverges. We thus obtain

$$S^{\text{on-shell}} = -\frac{dV_dR^{d-1}}{\kappa} \,. \tag{1.28}$$

Perturbative solutions We focus on the 1-instanton sector and thus assume the potential has the following form

$$V_{\rm np}(\chi) = \lambda \cos(\chi f_\alpha)\,, \tag{1.29}$$

with $\lambda \ll 1$. We look for perturbative solutions in λ of the form

$$\begin{aligned}\chi(\tau) &= \chi_{(0)}(\tau) + \lambda\chi_{(1)}(\tau) + \lambda^2\chi_{(2)}(\tau) + \dots \\ \alpha(\tau) &= a_{(0)}(\tau) + \lambda a_{(1)}(\tau) + \lambda^2 a_{(2)}(\tau) + \dots\end{aligned} \tag{1.30}$$

We assume charged solution

$$\begin{aligned}Q &= \int_{S^d} \star H = \int (\star)^2 d\chi \\ &= Q_{(0)} + \lambda Q_{(1)} + \lambda^2 Q_{(2)} + \dots \\ &= \int_{S^d} (\star)^2 d\chi_{(0)} + \lambda \int_{S^d} (\star)^2 d\chi_{(1)} + \lambda^2 \int_{S^d} (\star)^2 d\chi_{(2)}\,.\end{aligned} \tag{1.31}$$

There are two observations. First, the trivial set of solutions discussed previously have vanishing axion charge. Second, these solutions have running charges a priori. This can be seen by integrating the first equation on the compact slices of (1.22) yielding

$$\partial_\tau Q + (-1)^d V_d\, a^d \partial_\chi V_{\rm np} = 0\,. \tag{1.32}$$

As one can observe the non-perturbative potential has the effect of non-conservation of the axionic charge. We now wish to describe a perturbative way to obtain the solution. We still discuss k=1 in the $N = 1$ gauge. At order λ^0, we obtain

$$\begin{aligned}\partial_\tau \left(a_{(0)}^d \partial_\tau \chi_{(0)}\right) &= 0\,, \\ \frac{d(d-1)}{a_{(0)}^2}\left(a_{(0)}'^2 - 1\right) + 2\kappa\left(\frac{(\partial_\tau\chi_{(0)})^2}{2} + \frac{\Lambda}{\kappa}\right) &= 0\,.\end{aligned} \tag{1.33}$$

As we can observe the charge is constant at order λ^0 so $Q_{(0)} = \text{const}$. These equations become

$$\begin{aligned}a_{(0)}^d \partial_\tau \chi_{(0)} &= (-1)^d Q\,. \\ a_{(0)}'^2 &= 1 - \frac{2\kappa}{(a_0)^2 d(d-1)}\left(\frac{Q^2}{(V_d)^2 a_{(0)}^{2d}} + \frac{\Lambda}{\kappa}\right) \\ &= 1 - \frac{1}{a_{(0)}^2 H_{dS}^2} - \frac{2\kappa}{d(d-1)}\frac{Q^2}{(V_d)^2 a_{(0)}^{2d+2}}\,.\end{aligned} \tag{1.34}$$

Once $a_{(0)}$ is known, $\chi_{(0)}$ is also known.

At order λ^1, one obtains

$$\begin{aligned}\partial_\tau\left(a_{(0)}^d\partial_\tau\chi_{(1)} + a_{(1)}^d\partial_\tau\chi_{(0)}\right) - a_{(0)}^d \lambda f_\alpha \sin(\chi_{(0)}) &= 0\,. \\ \frac{d(d-1)}{a_{(0)}^2}\left(a_{(0)}'a_{(1)}' - \frac{a_{(1)}(a_{(0)}')^2}{a_{(0)}} - \frac{a_{(1)}k}{a_{(0)}}\right) + 2\kappa\left(\partial_\tau\chi_{(0)}\partial_\tau\chi_{(1)} + \frac{\cos(\chi_{(0)}f_\alpha)}{2}\right) &= 0\,.\end{aligned} \tag{1.35}$$

Now the question is whether or not we can solve analytically these set of perturbative equations.

2 Dynamical system analysis

2.1 First order formalism in expansion normalized variables

Recall the system of differential equations in the $N = 1$ gauge (1.22). We would like to write it as a first order autonomous system.PG: Actually, in litterature, such problem is common and the usual thing to do is to view the Fridmann equation as a constraint equation and to use EN (Expansion normalized) variables. The following references will be useful in our setup:
https://arxiv.org/abs/1712.03107
https://arxiv.org/pdf/0906.0396
https://arxiv.org/pdf/astro-ph/0004138
https://arxiv.org/pdf/1108.4712
https://arxiv.org/pdf/1401.1959
The difference being that they work in Lorentzian spacetimes while we work in euclidean spacetimes
We define the EN variables

$$x = \sqrt{\frac{\kappa}{d(d-1)}}\frac{\chi'}{H} \quad , \quad y = \sqrt{\frac{2\kappa V_{\text{eff}}}{d(d-1)}}\frac{1}{H}\,, \tag{2.1}$$

H is the Hubble constant defined as $H = \frac{a'}{a}$. This needs to assume that $V_{\text{eff}}(\chi) = V_{np(\chi)} + \frac{\Lambda}{\kappa}$ is positive. The Friedman equation i.e the second equation of (1.22) becomes simply a constraint

$$1 - \frac{k}{H^2a^2} + x^2 + y^2 = 0\,. \tag{2.2}$$

PG: In the litterature, this is usually the inverse$1 = x^2 + y^2 + ...$ This is due to the fact that we work in euclidean time. We denote $\Omega_k = \frac{k}{H^2a^2}$ and $\Omega_\chi = x^2 + y^2$. The former should be viewed as a Casimir energy of the slices and the latter corresponds to the euclidean total energy density of the axion field. The constraint becomes

$$1 - \Omega_k + \Omega_\chi = 0\,. \tag{2.3}$$

This is already interesting to remark that this constraint cannot be satisfied in the case where the universe is open i.e $k \in \{0, -1\}$ as the euclidean energy density of the axion is manifestly positive (of course providing that x, y are real and well defined)! The conclusion would have been crucially different in Lorenztian times. In fact, we can retrieve such a case by Wick rotating the Hubble constant and the euclidean time $H \to iH, \tau = it$. In Lorentzian, the system would have been

$$1 = \Omega_k + y^2 - x^2 \longrightarrow \text{Lorentzian constraint}\,. \tag{2.4}$$

The fact that the sign of x is opposite to what we usually see is due to the fact that we deal with an axion field which has a negative kinetic term.

Let us comeback to (2.3). This equation describes an hyperboloid in the $\left(\sqrt{\Omega_\chi}, \sqrt{\Omega_k}\right)$ plane. (In Lorentzian, it describes a circle!). This determines algebraically Ω_k as a function of (x, y).

$$\Omega_k(x, y) = 1 + x^2 + y^2\,. \tag{2.5}$$

Dividing by $d(d-1)H^2$ the third equation of (1.22) which is known to be the accelerating equation, one obtains

$$\frac{(d-2)}{d}\Omega_k - \frac{(d-2)}{d} - \frac{2}{d}\frac{a''}{a}\frac{1}{H^2} + x^2 - y^2 = 0\,. \tag{2.6}$$

We also have the identity $\frac{H'}{H^2} = \frac{aa''-a'^2}{H^2a^2} = \frac{a''}{aH^2} - 1$ which plugging back into the previous equation determines $\frac{H'}{H^2}$ in terms of the rest

$$\frac{(d-2)}{d}\Omega_k - \frac{(d-2)}{d} - \frac{2}{d} - \frac{2}{d}\frac{H'}{H^2} + x^2 - y^2 = 0\,. \tag{2.7}$$

Hence,

$$\frac{H'}{H^2} = \frac{d}{2}\left(x^2 - y^2 - 1 + \frac{(d-2)}{d}\Omega_k(x,y)\right) . \tag{2.8}$$

From the eom of the axion, we obtain

$$\frac{\chi''}{H^2} + \frac{d\chi'}{H} + \frac{\partial_\chi V_{\text{eff}}}{H^2} = 0\,. \tag{2.9}$$

Now we will express the equations of x and y in terms of the flow parameter $d\eta = Hd\tau = d(\log a)$. It suffices to take derivatives wrt the conformal time of the first order variables x, y.

$$\begin{aligned}\frac{dx}{d\eta} &= \frac{1}{H}\frac{dx}{d\tau} = -\frac{H'}{H^2}x + \frac{\chi''}{H^2}\sqrt{\frac{\kappa}{d(d-1)}} \\ &= x\frac{d}{2}\left(y^2 + 1 - x^2 - \frac{d-2}{d}\Omega_k(x,y)\right) - \sqrt{\frac{\kappa}{d(d-1)}}\frac{d\chi'}{H} - \frac{\partial_\chi V_{\text{eff}}}{H^2}\sqrt{\frac{\kappa}{d(d-1)}}\,.\end{aligned} \tag{2.10}$$

Observe that it is not an autonomous system as in our specific case $\partial_\chi V_{\text{eff}}$ depends on χ. A convenient way to deal with that is to define a third variable $\lambda = -\frac{\partial_\chi V_{\text{eff}}}{\sqrt{\kappa}V_{\text{eff}}}$. The previous equation can be defined in terms of y, λ

$$\frac{dx}{d\eta} = x\frac{d}{2}\left(y^2 + 1 - x^2 - \frac{(d-2)}{d}\Omega_k(x,y)\right) - dx + \frac{\sqrt{d(d-1)}}{2}\lambda y^2\,. \tag{2.11}$$

There is a historical reason why λ was introduced in cosmology. Initially, people were dealing with exponential potential for the inflaton for which λ is trivially a constant. In our case, λ is not a constant but there is a trick. It solves the following equation

$$\frac{d\lambda}{d\eta} = -\sqrt{d(d-1)}(\Gamma - 1)\lambda^2 x\,, \tag{2.12}$$

where $\Gamma = \frac{V_{\text{eff}}\partial_\chi^2 V_{\text{eff}}}{(\partial_\chi V_{\text{eff}})^2}$. In general, it is still a non autonomous system but in our specific case for which we have a cos potential we can actually try to express Γ in terms of λ so that to obtain a first order autonomous system of dimension 3. We will comeback to this equation latter. Let's now derive the equation for y. We use the same strategy i.e taking derivatives wrt the conformal time η

$$\begin{aligned}\frac{dy}{d\eta} &= \frac{1}{H}\frac{dy}{d\tau} = -\frac{H'}{H^2}y + \frac{1}{H^2}\sqrt{\frac{2\kappa}{d(d-1)}}\frac{\chi'\,\partial_\chi V_{\text{eff}}}{2\sqrt{V_{\text{eff}}}}\,. \\ &= y\frac{d}{2}\left(y^2 + 1 - x^2 - \frac{(d-2)}{d}\Omega_k(x,y)\right) - \frac{\sqrt{d(d-1)}}{2}xy\lambda\,.\end{aligned} \tag{2.13}$$

The three dimensional autonomous system reads simply

$$\begin{aligned}\frac{dx}{d\eta} &= x\frac{d}{2}\left(y^2 + 1 - x^2 - \frac{(d-2)}{d}\Omega_k(x,y)\right) - dx + \frac{\sqrt{d(d-1)}}{2}\lambda y^2\,, \\ \frac{dy}{d\eta} &= y\frac{d}{2}\left(y^2 + 1 - x^2 - \frac{(d-2)}{d}\Omega_k(x,y)\right) - \frac{\sqrt{d(d-1)}}{2}xy\lambda \\ \frac{d\lambda}{d\eta} &= -\sqrt{d(d-1)}(\Gamma(\lambda) - 1)\lambda^2 x\,.\end{aligned} \tag{2.14}$$

We now give specific examples for the potential for which we have $\Gamma(\lambda)$ which can easily be expressed in terms of the other variables. An interesting form of the potential is

$$V_{\text{eff}}(\chi) = V_0\cos^2(\chi f_\chi)\,. \tag{2.15}$$

We subsequently find

$$\begin{aligned}\lambda &= 2\frac{f_\chi}{\sqrt{\kappa}}\tan(\chi f_\chi)\,,\\ \Gamma(\lambda) &= \frac{1}{2} - \frac{2f_\chi^2}{\kappa}\frac{1}{\lambda^2}\,.\end{aligned} \tag{2.16}$$

Then the system becomes

$$\begin{aligned}\frac{dx}{d\eta} &= x\frac{d}{2}\left(y^2+1-x^2-\frac{(d-2)}{d}\Omega_k(x,y)\right) - dx + \frac{\sqrt{d(d-1)}}{2}\lambda y^2\,,\\ \frac{dy}{d\eta} &= y\frac{d}{2}\left(y^2+1-x^2-\frac{(d-2)}{d}\Omega_k(x,y)\right) - \frac{\sqrt{d(d-1)}}{2}xy\lambda\\ \frac{d\lambda}{d\eta} &= \sqrt{d(d-1)}\left(\frac{2f_\chi^2}{\kappa}+\frac{\lambda^2}{2}\right)x\,.\end{aligned} \tag{2.17}$$

In this case, it is relatively easy to discuss critical points. From the third equation, $x_\star = 0$ and then $y_\star = 0$. $\lambda_\star$ is arbitrary so the critical points correspond to the line $x = y = 0$. This needs nuances as when $y_\star = 0$, $V_{\text{eff}} = 0$ for which tan blows up at such a point so $\lambda_\star \to \infty$ is the only physical critical point that you can reach. They correspond exactly to the non-running flat solutions found discussed previously. This is very simple to see the line of critical points is unstable. We look at the linearized system $\lambda = \lambda_\star + \delta\lambda$, $x = \delta x$, $y = \delta y$, and one obtains

$$\frac{d\delta X}{d\eta} = \begin{pmatrix} -(d-1) & 0 & 0\\ 0 & 1 & 0\\ A_\star & 0 & 0\end{pmatrix}\delta X\,, \tag{2.18}$$

where $A_\star$ is a constant depending on the parameters. (This does not apply if $\lambda_\star = \infty$). We see that for $d > 1$, these are unstable critical points.

The problem of the EN variables introduced earlier is that they generically do not allow for wormhole solutions which are supposed to be $\mathbb{Z}_2$ so with initial conditions $\chi'(0) = a'(0) = 0$, $a(0) \neq 0$. Indeed, the change of variables x, y are singular at that point. We thus look for other variables which support wormhole solutions.

2.2 First order formalism in "wormhole" variables

Note that we can't use the first order formalism introduced in https://arxiv.org/pdf/2505.10703 for instance. In fact, they use $\phi' = i\chi'$ as the natural flow variable which breaks down in our case because precisely at the $\mathbb{Z}_2$ symmetric point for instance, the flow parameter degenerates. A convenient way to introduce wormhole-type of variables is to modify the EN variables by removing their singular part by removing H. We thus define

$$p = \sqrt{\frac{\kappa}{d(d-1)}}a\chi' \quad , \quad q = \sqrt{\frac{2\kappa V_{\text{eff}}}{d(d-1)}}a \quad , \quad u = aH = a'\,. \tag{2.19}$$

The constraint equation now becomes

$$u^2 + p^2 + q^2 = k = 1\,. \tag{2.20}$$

This is an algebraic system which defines S^3. The phase space is submanifold of S^3. Wormhole solutions are initially located at the north pole $q = 0$. Actually, this is important to notice that there are two branches $u = \sigma\sqrt{1-p^2-q^2}$ with $\sigma = \pm$. Performing the same steps as before but with the flow parameter $ds = \frac{d\tau}{a}$, one obtains

$$\begin{aligned}\frac{dp}{ds} &= \sigma\sqrt{1-p^2-q^2}\,p(1-d) + \frac{\sqrt{d(d-1)}}{2}\lambda q^2 \\ \frac{dq}{ds} &= \sigma\sqrt{1-p^2-q^2}\,q - pq\lambda\frac{\sqrt{d(d-1)}}{2} \\ \frac{d\lambda}{ds} &= -\sqrt{d(d-1)}\,(\Gamma-1)\,\lambda^2 x\,.\end{aligned} \tag{2.21}$$

The sigma sign defines contracting or expanding branches. The system is branchwise autonomous. Also, note that the square root has the bad behavior of breaking the Lipschitz assumption. This signals potential non uniqueness of the solution even for fixed branch. (This was actually to be expected as even in the abscence of a potential for $V_{\rm eff} = \text{const.}$, wormholes with infinite number of bubbles where found.) To obtain a wormhole solution, you shall glue a contracting branch for $s < 0$ to an expanding branch at $s > 0$. It might be possible to generate by this gluing procedure wormholes that are not $\mathbb{Z}_2$ symmetric.

Frieman Vaga variables For the conventions, we refer to https://arxiv.org/pdf/astro-ph/0004138. We assume our potential+cosmological constant is given by $V(\chi) = V_0 + V_1\cos\left(\frac{\chi}{f_\chi}\right)$. There are five possibilities:

- $V_1 > V_0 > 0$ and this potentially describes solutions interpolating between AdS and dS saddles.
- $V_0 > V_1 > 0$ and this describes dS saddles.
- $V_0 < V_1 < 0$ and this describes AdS saddles.
- $V_0 = V_1 < 0$ and this describes solutions interpolating between AdS and flat saddles.
- $V_0 = V_1 > 0$ and this describes solutions interpolating between dS and flat saddles.

We introduce the variables

$$J = \sqrt{\kappa}\chi' \quad , \quad I = \frac{\chi}{f_\chi} \quad , \quad H = \frac{a'}{a}. \quad , \quad ' = \frac{d}{d\tau}\,. \tag{2.22}$$

The equations of motion for the χ particle(see (1.22)) gives the first order equation for J

$$J' = -dJH + V_1\frac{\sqrt{\kappa}}{f_\chi}\sin(I)\,. \tag{2.23}$$

By definition of J and I, we have the relation

$$I' = \frac{1}{f_\chi\sqrt{\kappa}}J\,. \tag{2.24}$$

The accelerating equation of (1.22) can be recast in the form

$$-(d-1)(d-2)H^2 + \frac{k}{a^2}(d-1)(d-2) - 2\frac{a''}{a}(d-1) + J^2 - 2\kappa\left[V_0 + V_1\cos(I)\right] = 0\,. \tag{2.25}$$

Then, observe that $\frac{a''}{a} = H' + H^2$ so

$$-d(d-1)H^2 + \frac{k}{a^2}(d-1)(d-2) - 2H'(d-1) + J^2 - 2\kappa\left[V_0 + V_1\cos(I)\right] = 0\,. \tag{2.26}$$

To eliminate a, we then use the Friedman equation which states that

$$\frac{d(d-1)}{a^2}k = d(d-1)H^2 + J^2 + 2\kappa\left[V_0 + V_1\cos(I)\right] . \tag{2.27}$$

Injecting it into the previous equation and re-arranging, the final first order system reads

$$\begin{aligned} J' &= -dJH + V_1\frac{\sqrt{\kappa}}{f_\chi}\sin(I)\,, \\ I' &= \frac{1}{f_\chi\sqrt{\kappa}}J\,, \\ H' &= -H^2 + \frac{J^2}{d} - \frac{2\kappa}{d(d-1)}\left[V_0 + V_1\cos(I)\right] . \end{aligned} \tag{2.28}$$

To only deal with dimensionless quantities, we first introduce $M^2 = \kappa V_1$ which has dimension inverse length scare. $\alpha = f_\chi\sqrt{\kappa}$ is dimensionless. We define rescaled dimensionless quantities

$$v_0 = \frac{V_0}{M^2},\ \mathcal{J} = \frac{J}{M},\ \mathcal{H} = \frac{H}{M},\ du = Md\tau\,. \tag{2.29}$$

Then, we easily find

$$\begin{aligned} \frac{d\mathcal{J}}{du} &= -d\mathcal{J}\mathcal{H} + \frac{\sin(I)}{\alpha}\,, \\ \frac{dI}{du} &= \frac{\mathcal{J}}{\alpha}\,, \\ \frac{d\mathcal{H}}{du} &= -\mathcal{H}^2 + \frac{\mathcal{J}^2}{d} - \frac{2}{d(d-1)}\left[v_0 + \cos(I)\right] . \end{aligned} \tag{2.30}$$

PG: I guess, there could be some cosmological bounds on the α. Ioannis, what is your thought on that ? Assuming $d > 1$ and that the Planck length-scale is much smaller than the axion length scale $\alpha \ll 1$.

Case 1: $v_0 < 1$. We will first focus in such a case. We first run a solution for $\alpha = 0.1$, $v_0 = 0.5$, $d = 3$.

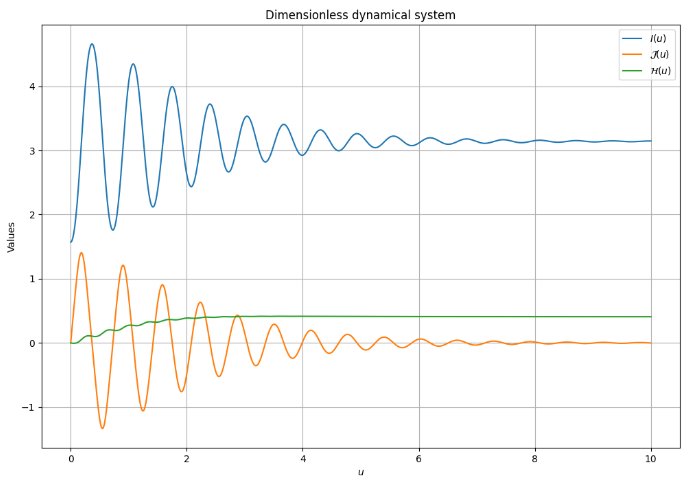

Figure 1: The autonomous system with initial conditions: $\mathcal{J} = 0$, $I = \frac{\pi}{2} \Rightarrow V(0) = V_1 v_0$, $\mathcal{H} = 0$

One can analytically derive that the critical points correspond to EAdS fixed points

$$\mathcal{J}_\star = 0 \quad , \quad I_\star = (2n+1)\pi \quad , \quad \mathcal{H}_\star = \pm\sqrt{\frac{2}{d(d-1)}(1-v_0)}\,. \tag{2.31}$$

Let's look at the linear stability. Denoting $\delta X = (\delta I, \delta\mathcal{J}, \delta\mathcal{H})$ One finds the following system

$$\frac{d\delta X}{d\eta} = \begin{pmatrix} 0 & \frac{1}{\alpha} & 0 \\ -\frac{1}{\alpha} & -d\mathcal{H}_\star & 0 \\ 0 & 0 & -2\mathcal{H}_\star \end{pmatrix} \delta X\,, \tag{2.32}$$

We first observe that $\lambda = -2\mathcal{H}_\star$ is an eigenvalue. First, requiring stability imposes the sign of $\mathcal{H}_\star > 0$. We thus focus in such case. The remaining two eigenvalues solve a second order equation

$$\lambda^2 + d\mathcal{H}_\star\lambda + \frac{1}{\alpha^2} = 0\,. \tag{2.33}$$

We can observe that the system is always stable. The product of the eigenvalues is positive and the sum is negative which imposes that both eigenvalues are negative. Nevertheless, the phase portrait is affected by the value of the parameters. If

$$\Delta < 0 \quad \text{i.e} \quad \frac{d}{(d-1)}(1-v_0) < \frac{2}{\alpha^2}\,, \tag{2.34}$$

the orbits will circle around the fixed point before reaching it. Otherwise, the orbits will directly flow to the fixed point. We can check that numerically by displaying the two type of solutions. The next step is to draw the phase portrait. Note that the χ particle describes a motion similar to a dissipative pendulum whose friction/anti-friction coefficient is time-dependent. This is natural to project the phase portrait into the plane $(I, \mathcal{J}) \sim (\chi, \Pi_\chi)$. We display the phase portrait in the two distinct regimes. As expected, in such regime, the motion of the χ particle is described by circular

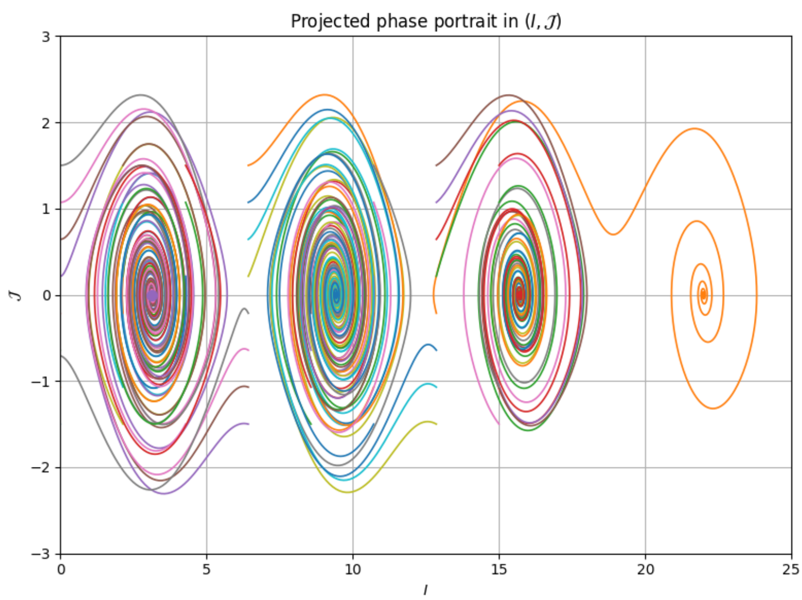

Figure 2: Phase portrait with parameter $v_0 = 0.5$, $\alpha = 0.2$, $d = 3$ and initial conditions $\mathcal{H}(0) = 0$. ~ 65 orbits are displayed.

orbits in the phase portrait. This phase portrait is similar to the one we know for a pendulum.

Nevertheless, it should be noted that it presents strking differences. Look at the orange curve to the right. It starts at a value of I in $[3\pi, 5\pi]$ but it ends in the critical point $\mathcal{I}_\star = 7\pi$. This hints toward the fact that the system is dominated by friction than anti-friction. If it was not the case, trajectories could flow away to infinity.

Conjecture: All the trajectories end at a critical point.

Holographic properties of the solutions. In this paragraph, we describe the holographic properties of the domain-wall solutions which asymptote to the EAdS fixed point. Near the AdS fixed point,

$$V(\chi) \to V_0 - V_1 < 0\,. \tag{2.35}$$

The spacetime asymptotes to an AdS geometry with length L_{AdS}. In terms of the dimensionless variable $\mathcal{H} = H/M$, one has

$$M L_{\text{AdS}} = \frac{1}{|\mathcal{H}_\star|} = \sqrt{\frac{d(d-1)}{2(1-v_0)}}\,. \tag{2.36}$$

Equivalently,

$$L_{\text{AdS}} = \sqrt{\frac{d(d-1)}{2\kappa V_1(1-v_0)}}\,. \tag{2.37}$$

Expanding around the fixed point $\chi_\star = (2n+1)\pi f_\chi$, and writing $\delta\chi = \chi - \chi_\star$, one obtains

$$V(\chi) = V_0 - V_1 + \frac{V_1}{2f_\chi^2}(\delta\chi)^2 + \mathcal{O}((\delta\chi)^4)\,. \tag{2.38}$$

Because the Euclidean axion has the wrong-sign kinetic term, the mass parameter entering the standard holographic scalar equation is

$$m^2_{\text{holo}} = -m^2_{\text{axion}} = -\frac{V_1}{f_\chi^2} = -\frac{M^2}{\alpha^2}\,. \tag{2.39}$$

Thus

$$m^2_{\text{holo}} L^2_{\text{AdS}} = -\frac{d(d-1)}{2\alpha^2(1-v_0)} < 0\,. \tag{2.40}$$

In the standard quantization, provided the BF bound is satisfied, this corresponds to a relevant deformation with $\Delta_+ < d$. The BF bound requires

$$m^2_{\text{holo}} L^2_{\text{AdS}} \geq -\frac{d^2}{4}. \tag{2.41}$$

For sufficiently small $\alpha^2(1-v_0)$, the BF bound is violated.